

Data Ethics

Research Data Foundations Camp
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[Photo by Elena Mozhvilo, available on Unsplash](#)

What does data ethics mean to you?

How do ethical questions show up in your work?

Optional worksheet to follow along:

<https://bit.ly/worksheet2026>

Today

1. Defining and making meaning of data ethics through the research lifecycle
2. The CARE Framework
3. Considering generative AI
4. **Small groups:** Applying CARE to Case Studies
5. Reflections, Questions, and Wrap Up

Data are constructed - they are the products of biased human choices and perspectives.

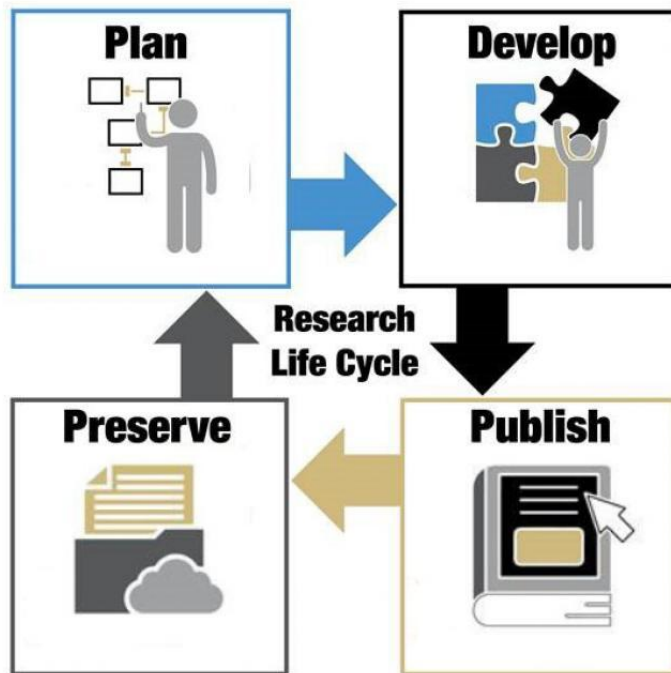
Data as Texts/Artifacts

- Data are not neutral or objective.
- Data do not represent a self-evident, discoverable Truth.

Further,

- Data are constructed, the products of biased human choices and perspectives.
- The interpretation, use, and analysis of data, like any text, is shaped by our biases and perspectives.

Thinking about Data Ethics through the Research Lifecycle



Our actions and responsibilities

- Who is responsible for data ethics?
- How do we make the most ethical choices in our research?
- CARE framework - paired with FAIR, a holistic look at machine actionability (FAIR) and social actionability (CARE)

CARE Framework

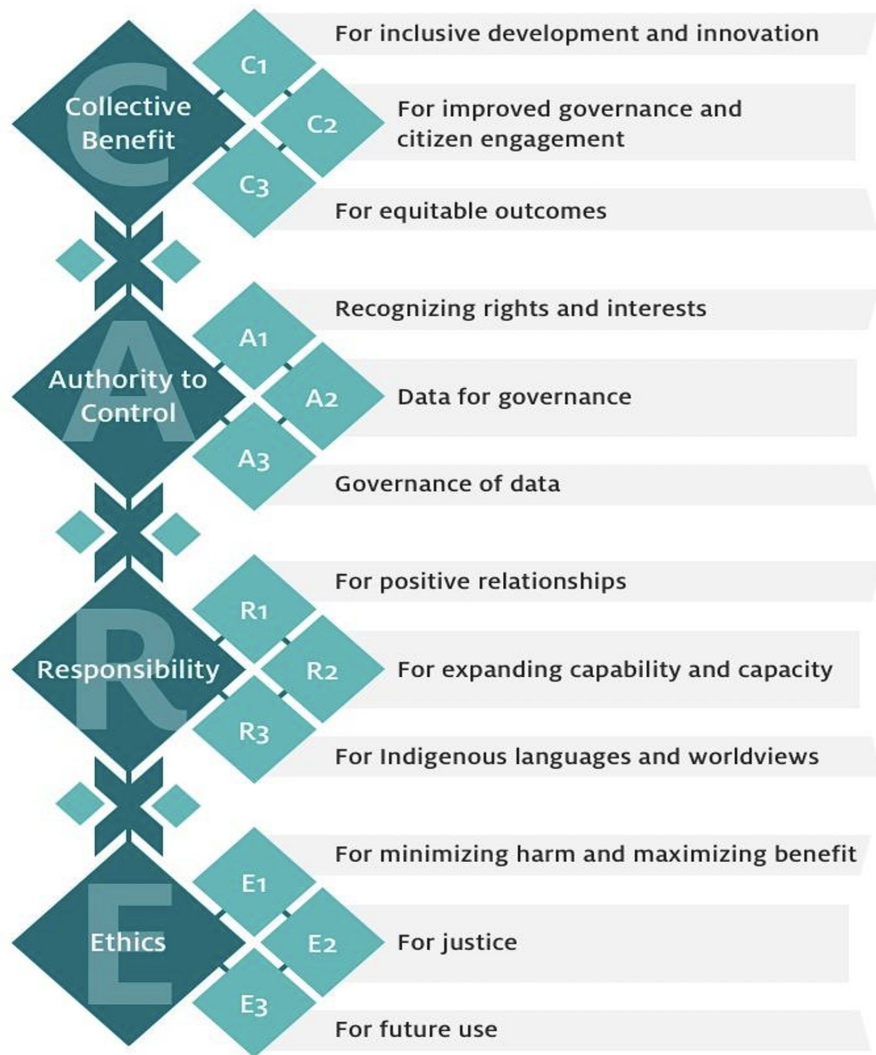
Principles for Indigenous Data Governance



*machine
actionability*

*social
actionability*

<https://www.go-fair.org/fair-principles/>
<https://www.gida-global.org/care>



<https://www.gida-global.org/care>

CARE: Principles to Questions

- From the perspective and ethical values of the involved community, what are the potential benefits and harms of collecting, using, or sharing these data now and in the future?
- Does the data play into harmful stereotypes and stigmas, or does it empower the community to assert their rights and interests?
- Are the provenances, purposes, and use limitations stated in the metadata or other documentation?
- Do metadata standards incorporate the (Indigenous) communities values and concepts?

So while AI is novel, it often seems to continue long-standing paradigms of technology in the service of capital ... scholarship suggests that the human harms documented in recent AI-driven initiatives are not merely “teething problems,” but part of a broader paradigm of capitalist and colonialist values at the core of our current economic and technological systems. (Munn, 2023)

**But ... AI
is
not novel**

So how can we
recognize ethical
questions around
power, trust, privacy &
security, and justice?

CARE for Generative AI

- **How is user input data treated?** Is it stored? Anonymized? Shared with other companies?
- Are these practices **transparent** and shared with users for **informed consent**?
- Is the output data **reproducible**? Was the model used documented?
- If I use a paid model of a genAI tool, how does that impact **access** and understanding of my work?
- Can I find **documentation** on the tool's corpus and training data? What are the checks against generating **misinformation**?
- What are **environmental impacts** of this tool vs. other methods?

Small Groups: Case Studies

(5 minute discussion)

Example Case Study: Cloning Dinosaurs



Your scientists were so preoccupied with whether or not they could that they didn't stop to think if they should.



What CARE-framed questions come up with this case?

Image credits: [Jurassic Park Wiki](#)

Breakout Rooms

<https://bit.ly/breakoutrooms2026>

**Back from small groups:
Discussing case studies
(10 minutes)**

Closing Reflections

- In what ways do you think *power* is reflected in your data?
- What data are you *missing*? Was it intentional? How might this missing data be beneficial?
- How might thinking about your data as a *text/artifact* change how you relate to, document, and share your work?

Thank you for joining us for Data Ethics!

References and further resources on following slides →

References (pg. 1)

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Munn, L. (2023) The five tests: designing and evaluating AI according to indigenous Māori principles. *AI & Society*.

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CARE Resources

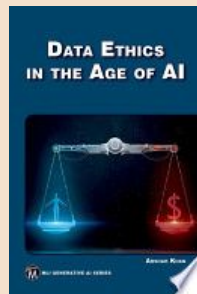
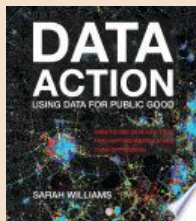
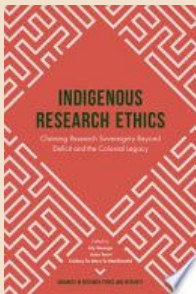
Publications identifying strategies, barriers, and needs in using CARE Principles for Indigenous Data Governance:

GENETIC AND BIOMEDICAL RESEARCH (2022)

GENOMIC RESEARCH (2022)

ECOLOGY AND BIODIVERSITY RESEARCH (2023)

Further Resources



[CU Boulder affiliates:
access these books
and more!](#)

**Federal Government
Data Strategy:
[Data Ethics Framework](#)
(2020)**

**Open Access:
[Shannon Vallor's An
Introduction to Data
Ethics](#)**